



Diet Analysis of Juvenile *Alosa pseudoharengus* (Alewife) and *Alosa sapidissima* (Shad) in Merrymeeting Bay

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Environmental Science and Policy

Thinking Matters

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Introduction

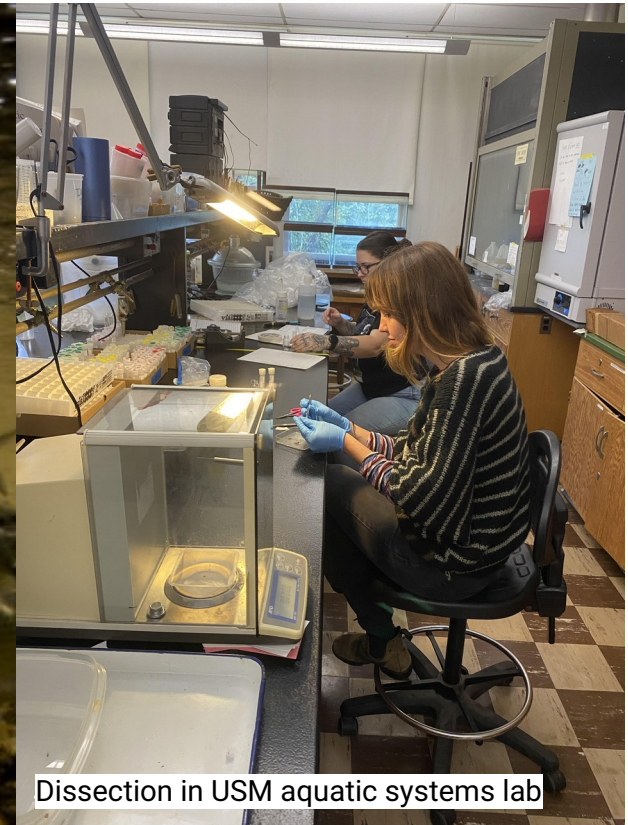
- **These species are vital ecological players.**
 - serve as key forage for larger predators
 - influence nutrient cycling
 - connect ecosystems across a broad geographic scale



I Research Background



© U.S. Fish and Wildlife Service | Alewife



Dissection in USM aquatic systems lab

I Research Background



Similar body shape and mouth structure suggest comparable feeding strategies and diet composition during juvenile stages, indicating potential overlap in trophic niche.

Knowledge Gap

- While both species are abundant in the same system, limited information exists on how juvenile alewife and shad partition resources
- Understanding whether they exhibit **trophic differentiation** (distinct dietary patterns) can inform ecosystem-based fisheries management

Questions

1. How might dietary patterns reflect species-specific life history strategies or habitat use?
2. Is there evidence of trophic overlap, or do these species show signs of niche differentiation?

Research Objectives

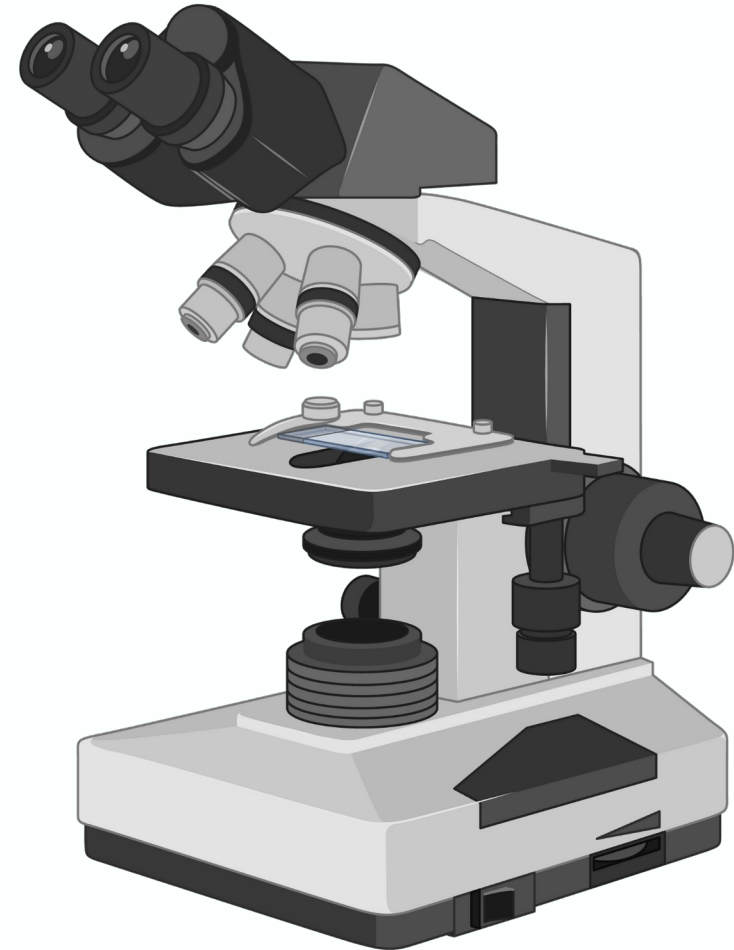
- I **To quantify and compare the stomach contents of juvenile** alewife and shad to assess dietary composition and diversity in Merrymeeting Bay.

- II **To explore trophic positioning and resource partitioning**, providing insight into long-term dietary patterns and ecological niche separation.

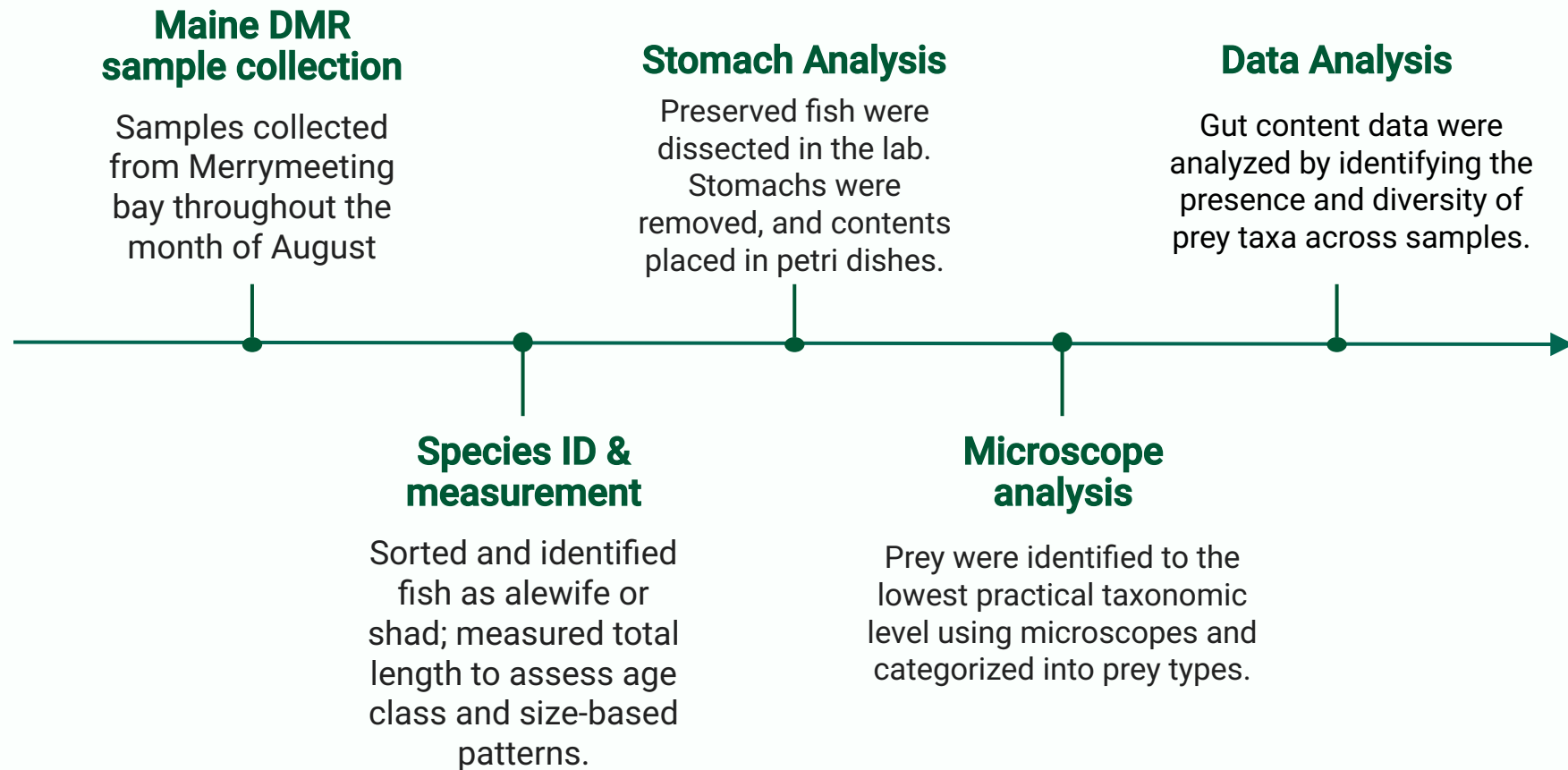
Research Methodology

This ongoing study combines **gut content analysis**, **stable isotope analysis** and **otolith aging** to examine habitat use patterns of juvenile alosine fish.

- **Diet analysis was conducted** using lab dissections and microscopy techniques.
- Prey items identified down to lowest practical taxonomic level



Timeline



I Research Background



Bosmina



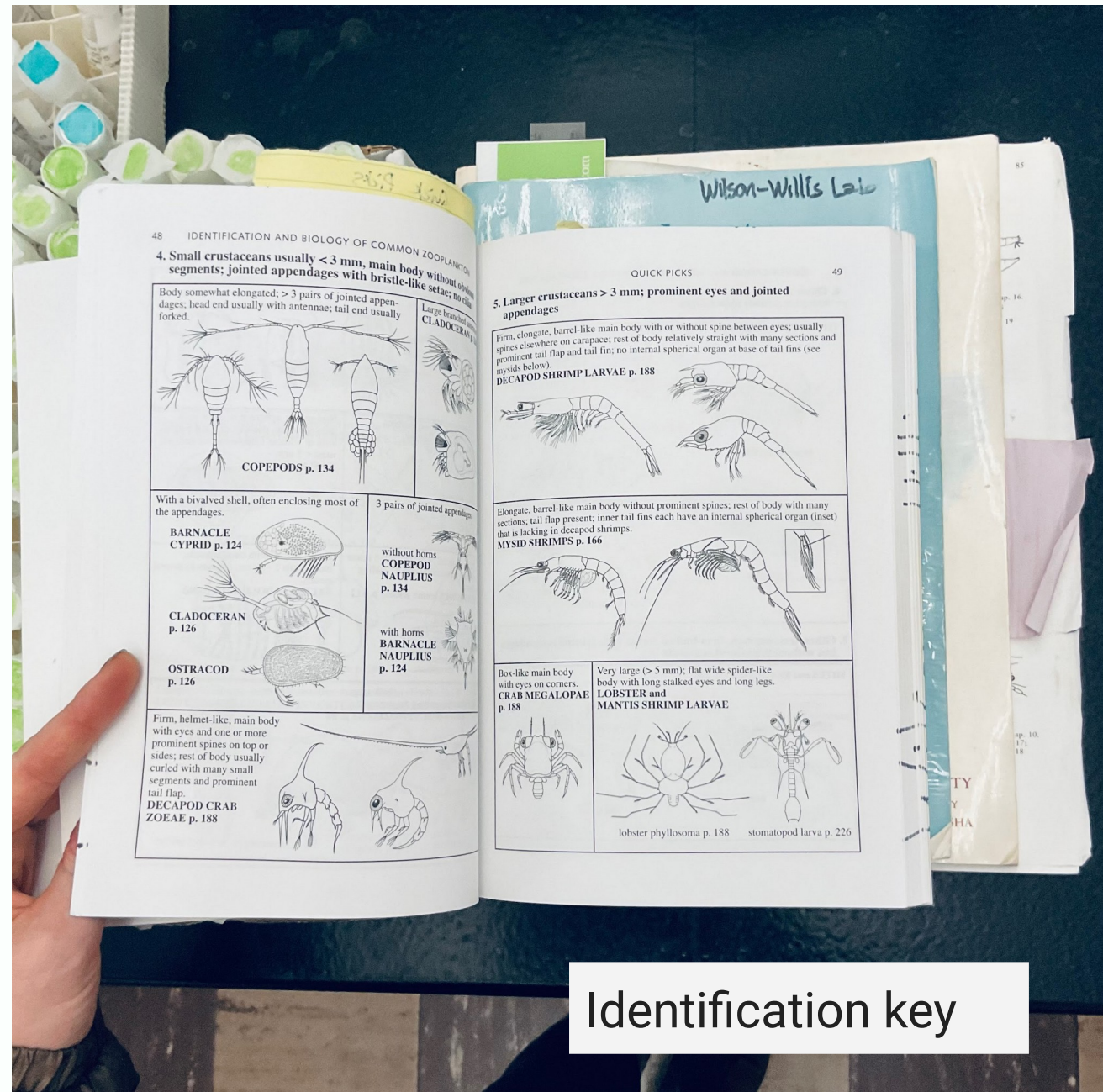
Ostracod



Copepod molt



Mysid Shrimp

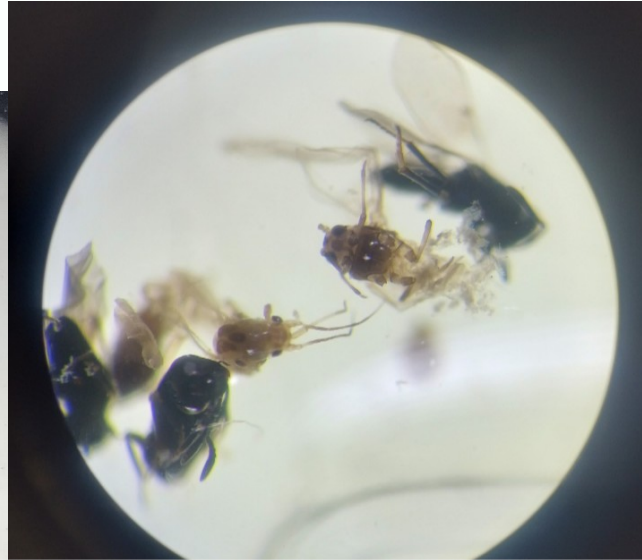


Identification key

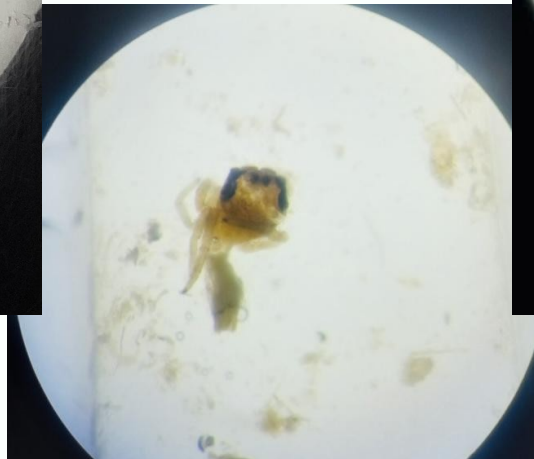
I Research Background



Shad stomach



Various winged insects found in a shad stomach



Partially digested prey item

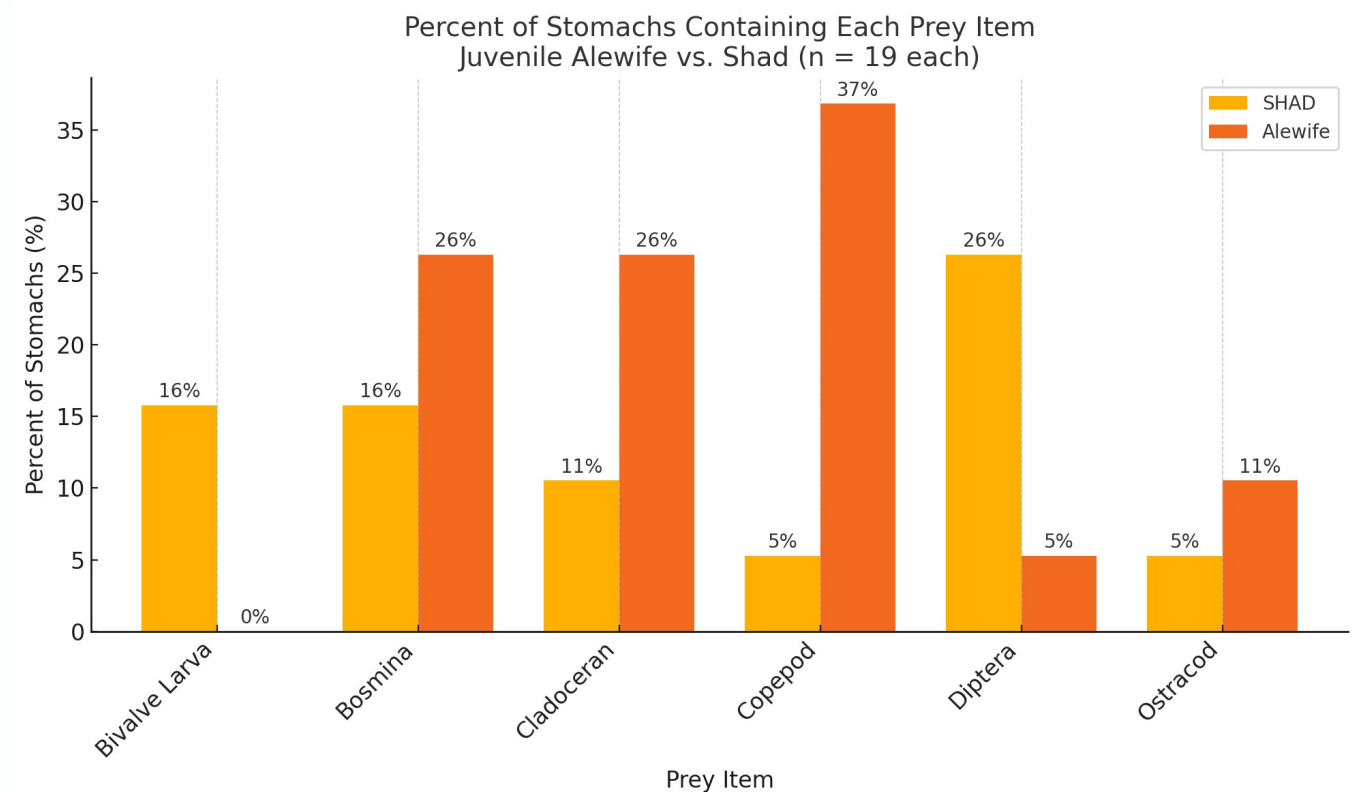


Detached limbs of prey item



Juvenile shad exhibit broader dietary breadth than alewife, indicating possible niche differentiation

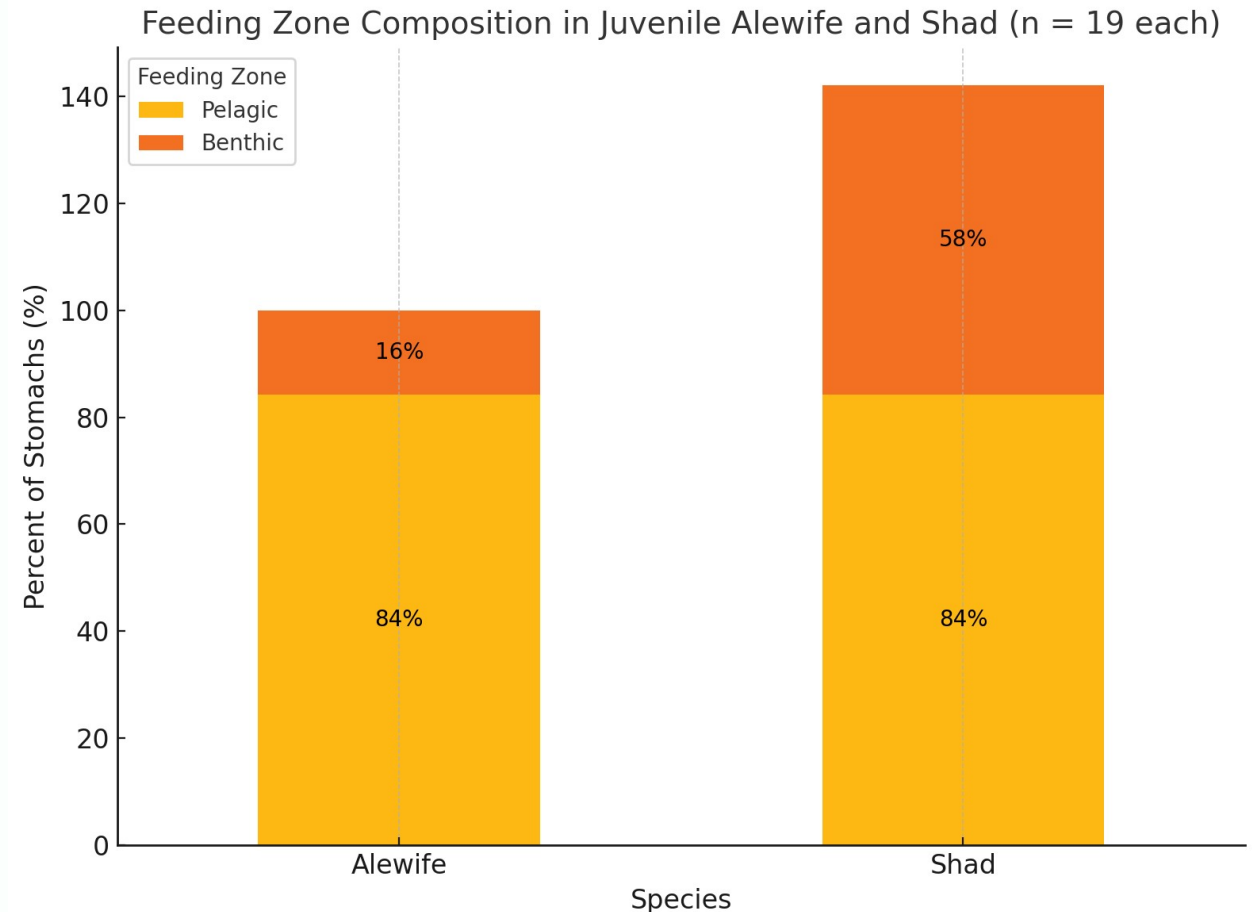
- **Alewife** consumed primarily **planktonic prey**, including copepods, cladocerans, and Bosmina.
- **Shad diets were more diverse**, featuring both planktonic items and benthic or shoreline-associated prey like **Diptera larvae** and **bivalve larvae**.





The presence of insect larvae and greater prey variety in Juvenile Shad suggests foraging in benthic zones

- These subtle patterns could indicate **trophic niche separation**, reducing potential competition between species co-occurring in the bay.



Summary of Results

- I **Juvenile alewife diets were dominated by pelagic prey** such as copepods, cladocerans, and Bosmina.
 - Indicates strong reliance on open-water zooplankton and pelagic foraging behavior.

- II **Juvenile shad consumed a broader range of prey**, including both pelagic and benthic organisms like Diptera and bivalve larvae.
 - Suggests more generalist feeding habits and foraging closer to the bottom of the bay

Discussion | Broader Implications

Ecosystem Function & Biodiversity

- Trophic differentiation between alewife and shad helps maintain species coexistence and ecosystem stability by reducing direct competition. Understanding these patterns contributes to a broader picture of how biodiversity is sustained in estuarine food webs.
- Protecting only open-water areas would support alewife, but not shad. To support both species, restoration efforts should also prioritize benthic habitats.

Discussion | Limitations

Sample Size

- Only 19 individuals per species were analyzed, which may limit the generalizability of the results and reduce statistical power for detecting more subtle patterns.

Temporal Scope

- All samples were collected within a single month, so the study does not account for potential seasonal shifts in diet or prey availability.

Looking ahead...

- Expand geographic coverage, sample size, and temporal scope
- Compare data with stable isotope analysis for a more long-term story of life strategies
- Examine overlap with additional species like blueback herring



Questions?

Acknowledgments

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